

10 The density of a sample of helium gas at the pressure of 100 kPa is 0.180 kg m^{-3} . The root-mean-square speed of the helium molecules is

A 41 m s^{-1}

B 561 m s^{-1}

C 1290 m s^{-1}

D 1685 m s^{-1}

11 A car tyre, initially at 20°C , has been inflated to a pressure of 160 kPa, as indicated by the